Scalability & Future Plan

Date	01 October 2026
Team ID	SWTID-2026-5883
Project Name	Pocket Smart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	1 Mark

Scalability & Future Plan

This plan describes how Pocket Smart AI can be extended for larger user bases, increased recommendation requests, additional data, stronger reliability and new planning categories. The current project includes Home Interior, Party and Jewelry planners, Gemini-based recommendations, user authentication/session features and recommendation history.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Dependence on external AI and product/service platforms	API availability, response time, rate limits or changes in external services may affect recommendation generation.	High
2	Increasing user/session and recommendation-history data	As users and saved recommendations increase, storage and retrieval requirements will grow.	Medium
3	Response time during high usage	Multiple simultaneous AI recommendation requests may increase processing time and affect user experience.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Web application and API-based planner requests.	Deploy multiple backend instances behind a load balancer; use asynchronous processing and request queuing where appropriate.
2	Data Storage	User/session information and recommendation history are maintained for personalization and review.	Move to scalable managed database/storage services, add indexing and archiving, and separate frequently accessed data from historical records.
3	Performance	AI-generated recommendations depend on model/API response time and external platform data.	Add caching for repeated queries, optimize prompts, use background jobs for longer tasks, and monitor API/model latency.
4	Security	Registration, login, JWT/session routes and CORS are included.	Strengthen authentication, secret management, access controls, rate limiting, input validation, audit logging and secure deployment practices.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Performance & Reliability: caching, API monitoring, stronger input validation, improved fallback recommendations and prompt optimization.	0–3 months	Faster and more consistent recommendations; improved reliability during normal usage.
Phase 2	Scalable Infrastructure: load balancing, managed database, asynchronous jobs and deployment automation.	3–6 months	Supports more concurrent users and larger recommendation workloads.
Phase 3	Expanded Intelligence: improved personalization, preference learning, richer multimodal support and additional budget-planning categories.	6–12 months	More relevant recommendations and broader usefulness across lifestyle planning needs.
Phase 4	Ecosystem Expansion: more product/service integrations, analytics dashboard, advanced recommendation history and broader platform availability.	12+ months	Wider coverage, stronger personalization and a more complete budget-assistance ecosystem.

Team Work Allocation

S.No.	Name	Assigned Work
1	Jashvini.K	AI scalability: Gemini integration optimization, prompt efficiency, recommendation caching and future personalization improvements.
2	Indhumathi.D	Backend scalability: API optimization, load handling, authentication/security improvements and scalable deployment.
3	Jayapriya Rajkumar	Frontend scalability: responsive UI improvements, faster result rendering, user experience and new planner interfaces.
4	Jonal.A	Integration scalability: external platform/API expansion, data integration, fallback mechanisms and additional planning categories.
5	Kavibharathi.K	Testing, performance monitoring and documentation: load testing, security checks, roadmap tracking and final scalability documentation.

The document follows the supplied Scalability & Future Plan template, which organizes the submission into current limitations, a four-part scalability plan (User Load, Data Storage, Performance, Security), and a future roadmap. ■filecite■turn4file0■L7-L24■